

The Synthesis of "Optimum" Transient Response: Criteria and Standard Forms

DUNSTAN GRAHAM
MEMBER AIEE

R. C. LATHROP
ASSOCIATE MEMBER AIEE

Synopsis: Methods for synthesizing servomechanisms are reviewed, and it is pointed out that stability criteria, frequency, and root-locus methods reduce to conditions on the transfer-function constants. These may be made mathematically specific following Whiteley's suggestion of "standard forms." Eight mathematical criteria for optimum transient responses are critically examined, and the clear superiority of the minimum integral of time-multiplied absolute-value of error is demonstrated. The application of this criterion results in the selection of standard forms, which are presented in tables.

TRANSIENT BEHAVIOR is an attribute of many measurement, control, and communication devices. Engineering design of such physical systems often involves the choice of design variables which will insure optimum transient behavior. Mathematical techniques applicable to this engineering design problem have been rapidly developed and extended during the past ten years. The lack of a practical mathematical definition of optimum transient behavior, however, has forced empiricism into the design procedures, and the procedures themselves, for the most part, are esoteric and laborious. Development of a unitary figure of merit for transient behavior and the tabulation of optimum system characteristics would be valuable to the designer.

Mathematical methods applied to the problem of optimizing the transient behavior of physical systems depend uniformly on the concept of the response of a

describes the physical system. A simple positioning servomechanism, its linear second-order equation, and its transfer function are illustrated in Fig. 2. The transfer functions of the linear systems considered in this paper are always reduced to the normalized form

$$\frac{C(s)}{R(s)} = \frac{p_m s^m + \dots + p_2 s^2 + p_1 s + p_0}{s^n + q_{n-1} s^{n-1} + \dots + q_2 s^2 + q_1 s + 1} \quad (1)$$

Procedures for deriving the normalized transfer function from the system differential equation are outlined in Appendix I.

Three distinct problems occur in the application of differential equations and operational mathematics to physical systems. These problems have been named the analysis, instrument, and synthesis problems.

1. The analysis problem is: Given the input and the mathematical description of the system, find the output.
2. The instrument problem is: Given the output and the mathematical description, find the input.
3. The synthesis problem is: Given the input and the desired output, determine the mathematical description.

It is clear that the synthesis problem is intimately related to engineering design. Typical inputs are often known and the "desired" output may be subject to specification. The mathematical description of the required system is a preliminary to its physical realization. Realization of the physical system is excluded from con-

phase margin, or on a transient basis in terms of intuitive concepts of rise time and overshoot.

Nomenclature

$r(t)$ = input to a transfer system
 $c(t)$ = output, or response, of a transfer system
 $e(t) = r(t) - c(t)$ = transfer system error
 $E(s) = R(s) - C(s)$ = Laplace transform of error
 $w(t)$ = transfer-system weighting function
 $W(s) = C(s)/R(s)$ = transfer function of a linear system
 $j = \sqrt{-1}$
 ω = angular frequency
 ω_0 = natural angular frequency
 λ = Laplace complex variable
 $s = \lambda/\omega_0$ = normalized Laplace variable
 t = time
 P_i, Q_i = transfer-function coefficients
 p_i, q_i = normalized transfer-function coefficients
 $a_i, b, c, d, \dots$ = normalized transfer-function coefficients
 ζ = damping ratio
 $\alpha + j\beta$ = a root of a transfer-system characteristic equation
 C_i = error coefficient

Synthesis Methods

The first and most obvious condition on the output is stability. The output of a duplicator is unequivocally desired to be stable. Fortunately, stability can be precisely defined, and conditions for stability can be rather easily calculated. There are certain conditions on the coefficients of the characteristic equation which must be satisfied in order to insure that the system is stable.

The functional relations between the coefficients defined by the Routh-Hurwitz criterion must hold. These test functions are tabulated for the various orders of normalized characteristic equations through the eighth order in Table I. Theoretically, $n-1$ test functions have to be applied to each characteristic equation.